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Review -1

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GraphGuard: An Explainable Graph Neural Network for Intelligent Intrusion Detection

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ABSTRACT:

The increasing interconnection of enterprise networks, cloud services, and Internet of Things devices has resulted in a growing number of sophisticated cyber attacks. Conventional Intrusion Detection Systems (IDS) generally analyze network traffic as independent records and may therefore fail to capture the relationships and communication patterns between network entities. This limitation can reduce the ability to detect coordinated, evolving, and previously unseen attacks while also generating false alarms. This project proposes GraphGuard, an explainable Graph Neural Network (GNN)-based framework for intelligent intrusion detection. The proposed system represents network communication as a graph in which network devices are represented as nodes and communication flows are represented as edges. A GNN can then learn both traffic characteristics and structural relationships between communicating entities. An Adaptive Graph Construction (AGC) mechanism is proposed to dynamically update graph relationships according to changing communication behavior. In addition, Explainable Artificial Intelligence (XAI) techniques are incorporated to provide interpretable explanations for detected attacks. A Threat Propagation Score (TPS) is proposed to estimate the potential severity and spread of detected threats and assist in alert prioritization. The framework will be evaluated using benchmark datasets including CICIDS2017 and UNSW-NB15, with comparisons against conventional machine-learning and graph-based intrusion detection approaches.

KEYWORDS: Graph Neural Networks, Intrusion Detection System, Explainable Artificial Intelligence, Adaptive Graph Construction, Threat Propagation, Cybersecurity, Network Security, Deep Learning.

[1] INTRODUCTION:

The rapid growth of interconnected computer networks, cloud platforms, IoT devices, and distributed services has significantly increased the attack surface of modern digital infrastructure. Intrusion Detection Systems play an important role in identifying malicious network activities and protecting systems from unauthorized access. Traditional machine-learning-based IDS approaches generally treat network traffic as independent feature vectors. Although such approaches can achieve good classification performance, they do not naturally represent the relationships between communicating hosts, services, and network flows.

Graph Neural Networks provide an alternative representation by modelling network communication as a graph. In this representation, devices or network entities can be represented as nodes while communication relationships are represented as edges. This allows the model to learn both individual traffic characteristics and structural relationships. Research on GNN-based IDS has demonstrated the potential of graph representations for network and host-based intrusion detection. GraphGuard extends this direction by combining graph-based intrusion detection with adaptive graph construction, explainability, and threat propagation analysis. The overall objective is to provide not only an attack prediction but also useful information about why the attack was detected and how significant its potential propagation may be.

[2] PROBLEM STATEMENT:

Conventional Intrusion Detection Systems (IDS) primarily analyze network traffic as independent records and therefore have limited ability to capture the relationships and communication patterns among network entities. This limitation can affect the detection of coordinated, evolving, and previously unseen cyber attacks and may result in a high number of false-positive alerts. Although Graph Neural Network (GNN)-based approaches can model network traffic as interconnected graphs, existing approaches may rely on relatively static graph representations and often provide limited interpretability of their predictions.

Therefore, there is a need for an intelligent intrusion detection framework that can adapt to changing network communication relationships, learn both traffic and graph-level patterns, and provide understandable explanations for its decisions. In addition, detected threats need to be prioritized according to their potential severity and propagation across connected network entities. The proposed GraphGuard framework addresses these challenges by combining Adaptive Graph Construction (AGC), GNN-based intrusion detection, Explainable Artificial Intelligence (XAI), and a Threat Propagation Score (TPS) to provide an adaptive, interpretable, and relationship-aware approach to network intrusion detection.

[3] OBJECTIVES:

The main objective of the proposed GraphGuard framework is to develop an intelligent and explainable Graph Neural Network (GNN)-based intrusion detection system capable of identifying malicious network activities by learning both network traffic characteristics and relationships among communicating entities.

- To model network traffic as a graph and develop a GNN-based intrusion detection system by representing network entities as nodes and communication flows as edges, enabling the model to learn both traffic patterns and structural relationships for effective cyber attack detection.

- To implement Adaptive Graph Construction (AGC) that dynamically updates graph relationships based on changing network communication behaviour, ensuring the model adapts to evolving network environments.
- To integrate Explainable Artificial Intelligence (XAI) and Threat Propagation Score (TPS) in order to provide interpretable predictions and estimate the severity and propagation potential of detected threats for better alert prioritization.
- To evaluate the proposed GraphGuard framework using benchmark intrusion detection datasets such as CICIDS2017 and UNSW-NB15, ensuring reliable performance assessment.
- To compare GraphGuard with existing machine learning and graph-based intrusion detection approaches using standard evaluation metrics, and to provide an interpretable, relationship-aware system that assists security analysts in understanding and prioritizing cyber threats

[4] SCOPE OF THE PROJECT:

The scope of the proposed GraphGuard project is to develop a graph-based intelligent intrusion detection framework for identifying malicious activities in computer networks. The project focuses on using Graph Neural Networks (GNNs) to learn both network traffic characteristics and relationships between communicating network entities.

The scope includes the following:

1. Network Traffic Analysis and Graph Representation: Processing network traffic data obtained from benchmark intrusion detection datasets and modelling network entities as nodes and communication flows as edges to capture relationships within network traffic.

2. Adaptive Graph Construction: Developing an Adaptive Graph Construction (AGC) mechanism to update graph relationships according to changing communication behaviour.

3. GNN-Based Intrusion Detection and Attack Classification: Applying Graph Neural Network techniques to learn graph and traffic patterns and classify network activities as normal or malicious, including identification of different categories of cyber attacks supported by the selected datasets.

4. Explainable AI and Threat Propagation Analysis: Incorporating XAI techniques to provide understandable explanations for the predictions made by the intrusion detection model and introducing a Threat Propagation Score (TPS) to estimate the potential severity and propagation of detected threats and assist in alert prioritization.

5. Dataset-Based Evaluation and Performance Comparison: Evaluating the proposed framework using benchmark datasets such as CICIDS2017 and UNSW-NB15, and comparing GraphGuard with selected conventional machine-learning and graph-based intrusion detection approaches using standard evaluation metrics.

6. Result Visualization: Presenting detection results, attack categories, explanations, and threat-priority information in an understandable form for security analysis.

The project is primarily limited to network-level intrusion detection using benchmark datasets. It does not cover complete enterprise Security Operations Center (SOC) deployment, automated attack mitigation, or real-world network infrastructure integration.

[5] PROPOSED SYSTEM:

The proposed GraphGuard framework is an intelligent Graph Neural Network (GNN)-based Intrusion Detection System designed to detect malicious activities by considering both network traffic characteristics and relationships among communicating network entities. Network traffic obtained from benchmark datasets such as CICIDS2017 and UNSW-NB15 is first preprocessed to handle data quality, feature representation, and normalization requirements. The processed traffic is then represented as a graph, where network entities are modelled as nodes and their communication flows are represented as edges with relevant traffic features. This graph-based representation enables the system to learn structural and behavioural patterns that may not be captured when traffic records are treated independently.

To handle changes in network communication behaviour, GraphGuard incorporates an Adaptive Graph Construction (AGC) mechanism that dynamically updates graph relationships based on the evolving interactions between network entities. The resulting graph is processed using a GNN-based detection model to learn node, edge, and neighbourhood-level representations and classify network activities as normal or malicious, including different attack categories supported by the selected datasets. The system is further enhanced with Explainable Artificial Intelligence (XAI) techniques to identify and present the important features and graph relationships that contribute to each prediction, making the detection process more interpretable for security analysts.

In addition, GraphGuard introduces a Threat Propagation Score (TPS) to estimate the potential severity and propagation of detected threats across connected network entities and support alert prioritization. The final system presents the detected attack type, prediction confidence, relevant explanations, and threat-priority information through a result visualization layer. The performance of GraphGuard will be evaluated using standard metrics such as accuracy, precision, recall, F1-score, and false-positive rate, and compared with selected conventional machine-learning and graph-based intrusion detection approaches. Thus, the proposed system aims to provide an adaptive, explainable, relationship-aware, and threat-prioritized intrusion detection framework.

[6] LITERATURE SURVEY:

S.No.	Title / Journal or Conference / Publisher / Author(s) & Year	Methodology	Key Contribution	Limitations / Gap	Relevance to Project
1	"E-GraphSAGE: A Graph Neural Network Based Intrusion Detection System for IoT" — *NOMS 2022* — IEEE — Lo et al. (2022)	Represents IoT network communication as a graph and applies an edge-enhanced GraphSAGE model using node, edge, and neighbourhood information for intrusion detection.	Demonstrates that graph structure and communication relationships can improve intrusion detection compared with conventional traffic-based approaches.	The graph representation is not primarily designed for continuously changing network relationships. Integrated XAI and threat propagation prioritization are not addressed.	Provides the fundamental GNN-based graph representation and intrusion detection approach for GraphGuard.
2	"Anomal-E: A Self-Supervised Network Intrusion Detection System Based on Graph Neural Networks" — *Knowledge-Based Systems* — Elsevier — Caville et al. (2022)	Uses self-supervised graph representation learning to learn network-flow relationships and detect anomalous or intrusive activities.	Reduces dependence on fully labelled network traffic and demonstrates the effectiveness of GNNs for network anomaly detection.	Does not focus on dynamically modifying the graph according to changing network behaviour. Explainability and explicit threat propagation scoring are not integrated.	Supports GraphGuard's GNN-based detection and self-supervised learning direction.
3	"DIGNN-A: Real-Time Network Intrusion Detection with Integrated	Constructs a dynamic graph from network communication and uses	Addresses changing network topology and dynamic communication behaviour for	Focuses mainly on dynamic graph-based detection; a unified	Highly relevant to GraphGuard's Adaptive Graph Construction

	Neural Networks Based on Dynamic Graph" — *Computers, Materials & Continua* — Tech Science Press — Liu & Guo (2025)	neural-network-based graph learning for real-time intrusion detection.	real-time intrusion detection.	combination of dynamic graph construction, XAI, and threat propagation prioritization is not provided.	(AGC) and dynamic network modelling.
4	"Applying Self-Supervised Learning to Network Intrusion Detection for Network Flows with Graph Neural Network" — *Computer Networks* — Elsevier — Xu et al. (2024)	Applies self-supervised graph learning with a GNN-based graph encoder to learn representations from network flows and perform intrusion classification.	Reduces dependence on manually labelled network-flow data and supports multiclass intrusion detection.	Does not primarily address adaptive graph construction, prediction explainability, or threat propagation-based alert prioritization.	Supports GraphGuard's GNN representation learning and intrusion classification components.
5	"Self-Supervised Learning of Graph Representations for Network Intrusion Detection" — *Advances in Neural Information Processing Systems (NeurIPS)* — NeurIPS — Guerra et al. (2026)	Uses self-supervised learning to obtain graph representations for network intrusion detection, reducing reliance on labelled data.	Provides a recent graph representation learning approach for network intrusion detection and demonstrates the value of self-supervised graph learning.	Focuses primarily on representation learning and detection rather than integrating adaptive graph updates, XAI, and threat propagation analysis.	Provides a recent research direction for GraphGuard's graph representation and adaptive learning components.

6	"PPT-GNN: A Practical Pretrained Spatio-Temporal Graph Neural Network for Network Security" — *IEEE European Symposium on Security and Privacy Workshops (EuroS&PW)* — IEEE — Van Langendonck et al. (2025)	Uses a pretrained spatio-temporal GNN to learn network behaviour and capture temporal and structural patterns for network security applications.	Addresses practical issues such as limited labelled data, scalability, and generalization across network environments.	Pre-training and spatio-temporal learning are the primary focus; explicit XAI and threat propagation scoring are not central components.	Supports GraphGuard's objective of learning changing network behaviour using GNNs.
7	"Always Be Pre-Training: Representation Learning for Network Intrusion Detection with GNNs" — *2024 International Symposium on Quality Electronic Design (ISQED)* — IEEE — Gu et al. (2024)	Uses GNN-based representation learning with pre-training to learn useful network representations before intrusion-detection tasks.	Shows that pre-training can improve representation learning and reduce dependence on large amounts of labelled IDS data.	Does not provide an integrated adaptive graph construction mechanism or an XAI and threat-prioritization layer.	Supports GraphGuard's representation learning and adaptability to limited labelled network data.
8	"XG-BoT: An Explainable Deep Graph Neural Network for Botnet Detection and Forensics" —	Uses a deep GNN to model network communication for botnet detection and incorporates	Combines graph neural networks with explainability for cybersecurity and botnet detection/forensics.	Primarily focuses on botnet detection rather than a broader intrusion-detection	One of the most directly relevant papers for GraphGuard's GNN + XAI architecture.

	Internet of Things — Elsevier — Lo et al. (2023)	explainability to identify important factors behind predictions.		framework. Adaptive graph construction and threat propagation scoring are not integrated.	
9	"Interpreting Black-Box Models: A Review on Explainable Artificial Intelligence" — *Cognitive Computation* — Springer — Hassija et al. (2024)	Reviews major XAI techniques and approaches for interpreting black-box machine-learning and deep-learning models.	Establishes the importance of transparency, interpretability, and human-understandable explanations in AI-based decision systems.	It is a general XAI review and does not specifically design an explainable GNN-based network IDS.	Provides the XAI theoretical foundation for explaining GraphGuard's intrusion-detection predictions.
10	"Towards a Geometric Deep Learning-Based Cyber Security: Network System Intrusion Detection Using Graph Neural Networks" — *SECRYPT 2023* — SciTePress — Zaccagnino et al. (2023)	Applies geometric deep learning and GNN techniques to represent network systems and detect intrusion activities.	Demonstrates the applicability of geometric deep learning and GNNs to cybersecurity and network intrusion detection.	Focuses primarily on detection performance and graph-based learning rather than adaptive graph construction, explainability, and threat propagation.	Supports GraphGuard's graph-based cybersecurity and GNN intrusion-detection foundation.
11	"A Deep Learning Approach to Network Intrusion Detection" — *IEEE*	Uses deep-learning techniques to automatically learn representations from	Demonstrates that deep learning can effectively learn complex patterns for automated	Conventional traffic-feature representation does not inherently capture relationships	Useful as a deep-learning baseline for comparison with GraphGuard'

	Transactions on Emerging Topics in Computational Intelligence — IEEE — Shone et al. (2018)	network traffic features for intrusion detection.	network intrusion detection.	between communicating network entities. No graph modelling or integrated XAI is provided.	s relationship-aware GNN approach.
12	"Towards a Standard Feature Set for Network Intrusion Detection System Datasets" — *Mobile Networks and Applications* — Springer — Sarhan et al. (2022)	Analyses network intrusion datasets and investigates the consistency and usefulness of network-traffic features for NIDS evaluation.	Addresses inconsistencies in feature representation across intrusion-detection datasets and supports more meaningful dataset comparisons.	Does not propose a GNN-based detection architecture or adaptive graph-learning mechanism.	Relevant to feature selection, preprocessing, and evaluation of network IDS datasets used by GraphGuard.
13	"Graph Neural Networks: A Review of Methods and Applications" — *AI Open* — Elsevier — Zhou et al. (2020)	Reviews GNN architectures, graph representation learning, graph convolution, message passing, and major application areas.	Provides a comprehensive theoretical foundation for understanding how GNNs learn from graph structures and neighbourhood information.	General GNN survey rather than an intrusion-detection-specific solution. It does not address AGC, XAI, or TPS.	Supports the selection and theoretical design of GraphGuard's GNN architecture.
14	"A Comprehensive Survey on Graph Neural Networks" — *IEEE Transactions on Neural Networks and*	Surveys GNN architectures, learning paradigms, graph convolution methods, applications, and major	Provides a broad classification and understanding of GNN models and graph-learning techniques.	General-purpose survey; cybersecurity, adaptive graph construction, and threat propagation	Supports the GNN methodology, architecture selection, and theoretical background

	Learning Systems — IEEE — Wu et al. (2020)	research challenges.		are not the primary focus.	of GraphGuard.
15	"Machine Learning: Algorithms, Real-World Applications and Research Directions" — *SN Computer Science* — Springer — Sarker (2021)	Reviews major machine-learning algorithms, learning paradigms, and their real-world applications.	Provides a general foundation for machine-learning methodology, model selection, and practical AI applications.	Not specifically focused on graph neural networks, cybersecurity, or intrusion detection.	Provides the general ML foundation and baseline-comparison context for GraphGuard.

[6.1] FINDINGS IN LITERATURE SURVEY

The literature survey indicates that network intrusion detection has been extensively studied using machine learning, deep learning, Graph Neural Networks (GNNs), dynamic graph learning, self-supervised learning, and Explainable AI (XAI). Recent studies demonstrate that graph-based approaches can effectively capture relationships between communicating network entities, while self-supervised and pre-trained approaches improve representation learning with limited labelled data. However, the following key gaps were identified:

1. Limited adaptive graph construction: Existing GNN-based IDS approaches such as E-GraphSAGE and Anomal-E demonstrate the importance of graph-based network representation, while dynamic graph approaches address changing network structures. However, adaptive graph construction based on continuously evolving communication behaviour is not sufficiently integrated into a unified IDS framework.

2. Insufficient integration of traffic and structural relationships: Conventional deep-learning IDS approaches mainly process network traffic as feature vectors, while GNN approaches capture relationships between network entities. However, there is a need for a unified approach that effectively considers both traffic characteristics and changing structural relationships during intrusion detection.

3. Limited explainability in GNN-based IDS: Explainable approaches such as XG-BoT demonstrate the value of interpreting graph-based security predictions. However, explainability is not consistently integrated with adaptive GNN-based intrusion detection, particularly in terms of identifying the important network entities, traffic features, and graph relationships responsible for a prediction.

4. Lack of threat propagation and alert prioritization: Most existing intrusion detection approaches focus primarily on determining whether network activity is normal or malicious and, in some cases, identifying the attack category. They generally do not provide an explicit Threat Propagation Score (TPS) to estimate the potential severity or

spread of a detected threat across connected network entities and support alert prioritization.

5. Challenges in adapting to evolving attacks: Self-supervised and pre-trained GNN approaches improve representation learning and reduce dependence on labelled data. However, changing network communication patterns and evolving attack behaviours require detection frameworks that can adapt their graph representation and maintain effective detection performance as network relationships change.

6. Need for an integrated and interpretable framework: The reviewed studies individually address important aspects such as GNN-based intrusion detection, dynamic graphs, self-supervised learning, and XAI. However, these capabilities have not been sufficiently combined into a single framework that provides adaptive graph construction, relationship-aware intrusion detection, explainable predictions, and threat-prioritized alerts.

[7] METHODOLOGY:

1. **Literature Review and Baseline Selection:**
   Existing GNN-based, dynamic graph, self-supervised, and explainable intrusion detection approaches are reviewed to identify the research gaps addressed by GraphGuard. Suitable conventional machine-learning and GNN-based IDS models are selected as baselines for performance comparison.
2. **Dataset Acquisition and Preprocessing:**
   Benchmark network intrusion datasets, primarily CICIDS2017 and UNSW-NB15, are acquired and preprocessed. The preprocessing stage includes handling missing and duplicate records, removing irrelevant features, encoding categorical information, normalizing numerical features, and preparing attack labels. The processed data is divided into appropriate training, validation, and testing sets.
3. **Network Graph Construction:**
   The preprocessed network-flow records are transformed into graph representations. Network entities such as hosts or devices are represented as nodes, while communication flows between entities are represented as edges. Relevant traffic characteristics are associated with the corresponding nodes or edges to preserve both traffic-level and structural information.
4. **Adaptive Graph Construction:**
   An Adaptive Graph Construction (AGC) mechanism is designed to update the graph according to changing network communication behaviour. Changes in communication relationships and network activity are considered so that the graph representation can adapt instead of relying entirely on a fixed network topology. This component forms an important part of GraphGuard's ability to handle evolving network environments.
5. **GNN-Based Intrusion Detection:**
   The dynamically constructed graph is provided to a Graph Neural Network (GNN) model to learn representations from node features, edge features, and neighbourhood relationships. Suitable GNN architectures such as GraphSAGE, GAT, or GCN-based models can be evaluated, and the selected architecture is trained to classify network

activities as normal or malicious and identify attack categories supported by the datasets.

6. **Model Training and Attack Classification:**
   The GNN model is trained using the prepared training data with suitable optimization and class-balancing strategies where required. Validation data is used for model selection and hyperparameter tuning, while the test set is kept separate for final evaluation. The trained model generates predictions for normal and malicious traffic and, where applicable, individual attack categories.
7. **Explainable AI Analysis:**
   An Explainable Artificial Intelligence (XAI) module is incorporated to analyse the predictions generated by GraphGuard. Important traffic features, nodes, edges, or graph relationships contributing to a prediction are identified using suitable explanation techniques. These explanations are presented along with the prediction so that security analysts can better understand the basis of the detected intrusion.
8. **Threat Propagation Score Calculation:**
   A Threat Propagation Score (TPS) is designed to estimate the potential severity and propagation of a detected threat across connected network entities. Factors such as detection confidence, affected entities, suspicious communication relationships, and graph connectivity can be considered in calculating the score. The TPS is used primarily to prioritize security alerts rather than to claim exact prediction of future attack propagation.
9. **Evaluation and Benchmarking:**
   GraphGuard is evaluated using accuracy, precision, recall, F1-score, false-positive rate, confusion matrix, and other appropriate metrics. Performance is compared against selected conventional machine-learning and existing GNN-based IDS approaches. Separate analysis can also be performed for individual attack categories to determine the model's ability to detect different types of malicious activities.
10. **Ablation and Explainability Analysis:**
    Component-wise ablation studies are conducted to measure the contribution of major GraphGuard components, particularly **AGC, GNN detection, XAI, and TPS**. The performance of the complete framework is compared with versions in which individual components are removed or replaced. The generated explanations are also analysed to determine whether the identified features and graph relationships provide meaningful information for understanding and prioritizing intrusion alerts.

[8] SOFTWARE REQUIREMENTS:

Functional Requirements:

ID	Requirement
FR1	The system shall load and preprocess network intrusion datasets such as CICIDS2017 and UNSW-NB15, including data cleaning, feature handling, normalization, label preparation, and division into training, validation, and testing sets.

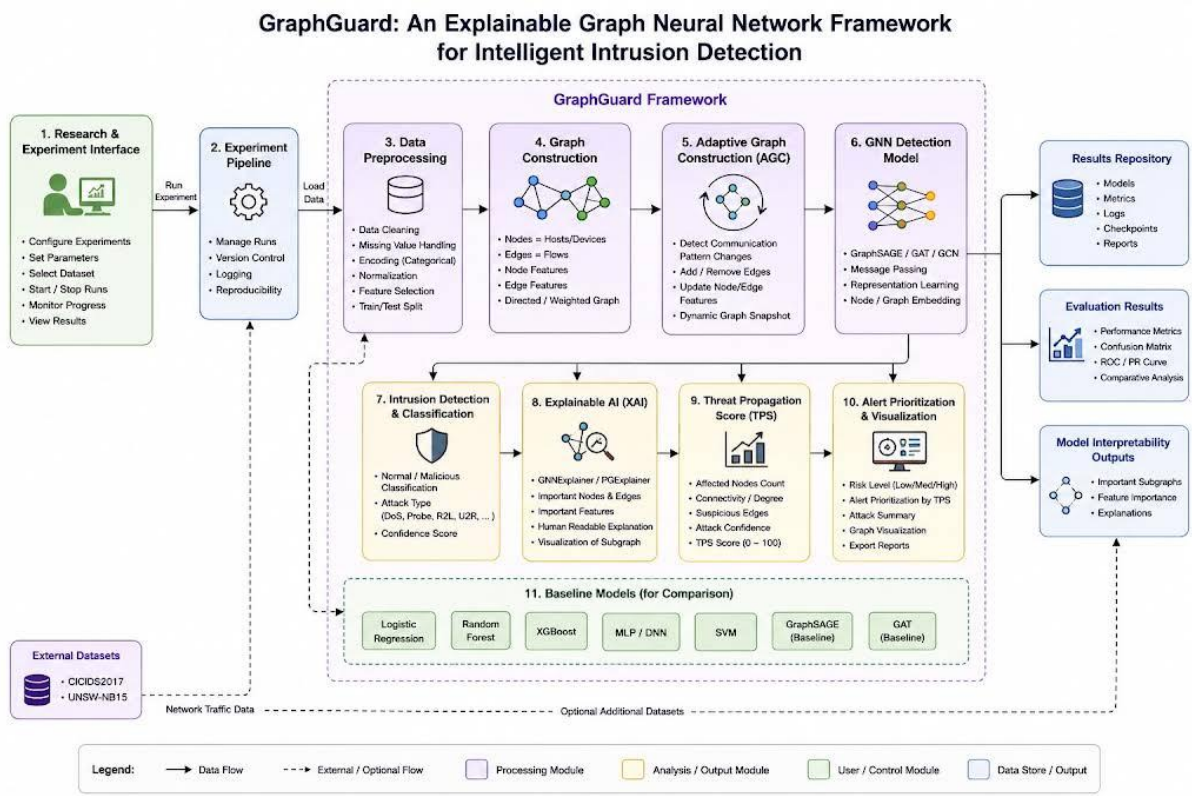
FR2	The system shall construct a graph representation of network traffic by modelling network entities as nodes, communication flows as edges, and associating relevant traffic features with the corresponding graph elements.
FR3	The system shall implement Adaptive Graph Construction (AGC) to update graph relationships according to changing network communication behaviour.
FR4	The system shall train a Graph Neural Network (GNN) to learn traffic and structural relationships and classify network activities as normal or malicious, including supported attack categories and prediction confidence.
FR5	The system shall implement an Explainable AI (XAI) module to identify important traffic features, nodes, edges, and graph relationships contributing to intrusion-detection predictions.
FR6	The system shall calculate a Threat Propagation Score (TPS) for detected threats and use it to prioritize security alerts, while displaying attack categories, explanations, confidence, and threat-priority information.
FR7	The system shall evaluate and benchmark GraphGuard using standard performance metrics and compare it with selected conventional machine-learning and GNN-based IDS approaches, including component-wise ablation analysis.

Non-Functional Requirements:

ID	Requirement
NFR1	**Accuracy & Reliability:** The system should provide reliable intrusion detection while minimizing incorrect classifications and false-positive alerts and should produce consistent results for the same input and configuration.
NFR2	**Performance & Efficiency:** The system should process network traffic, construct graphs, and perform GNN-based prediction within practically acceptable time and computational-resource requirements.
NFR3	**Scalability:** The system should be capable of handling large network intrusion datasets and graph representations without significant degradation in performance.
NFR4	**Adaptability:** The system should accommodate changes in network communication relationships through the Adaptive Graph Construction mechanism.
NFR5	**Explainability & Usability:** The system should provide understandable intrusion predictions, important contributing features or graph relationships, and threat-priority information in a clear form for security analysis.

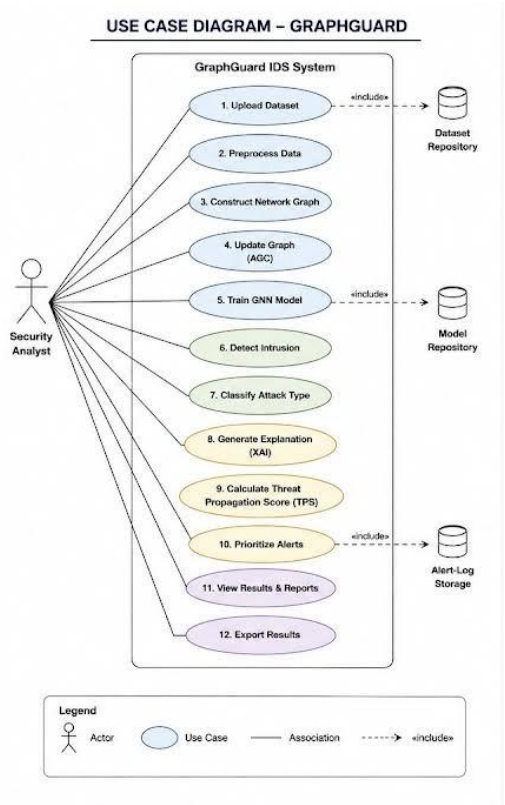
NFR6	**Maintainability & Extensibility:** The system should use modular components so that preprocessing, graph construction, GNN, XAI, TPS, datasets, and model architectures can be independently modified or extended.
NFR7	**Reproducibility & Security:** The experimental pipeline should maintain documented preprocessing, dataset splits, model configurations, and data integrity so that results can be reproduced reliably.

[9] SYSTEM ARCHITECTURE:

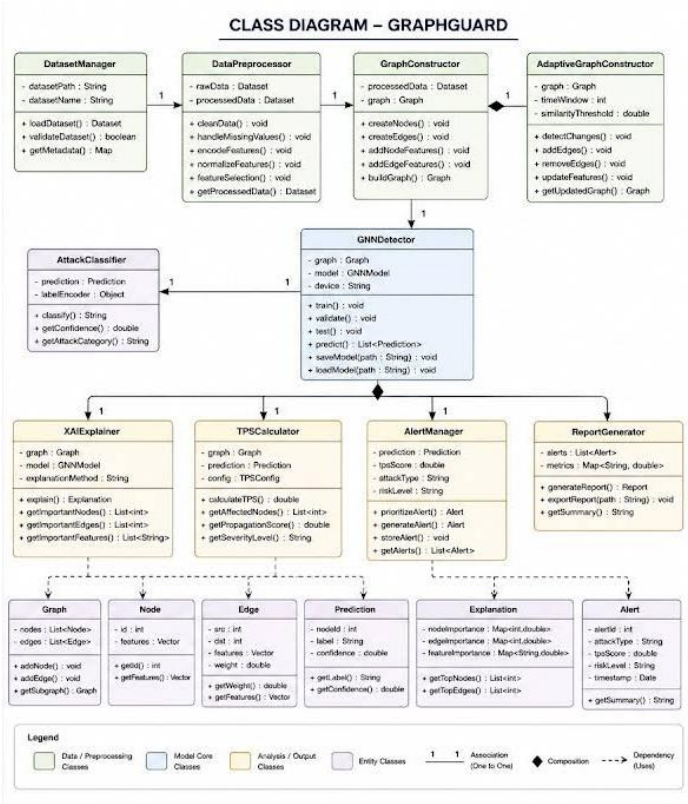


[10] UML DIAGRAM:

Use Case Diagram:



Class Diagram:



SUMMARY:

GraphGuard is an explainable Graph Neural Network (GNN)-based framework designed to improve intrusion detection in modern enterprise, cloud, and Internet of Things (IoT) networks.

Unlike conventional Intrusion Detection Systems (IDS) that primarily analyze network traffic as independent records, GraphGuard models communication as a graph, where network devices are represented as nodes and communication flows as edges. This enables the system to capture both traffic characteristics and relationships between network entities, helping identify coordinated and evolving cyber attacks.

The framework introduces an Adaptive Graph Construction (AGC) mechanism to dynamically update network relationships according to changing communication behavior. Explainable Artificial Intelligence (XAI) techniques are incorporated to provide understandable explanations for detected threats, improving the transparency and trustworthiness of the detection process. In addition, the proposed Threat Propagation Score (TPS) estimates the potential severity and spread of detected attacks, supporting effective alert prioritization.

GraphGuard will be evaluated using benchmark intrusion detection datasets, including CICIDS2017 and UNSW-NB15, and compared with conventional machine-learning and graph-based approaches. The proposed framework aims to achieve accurate, adaptive, and explainable intrusion detection while reducing false alarms and improving the identification of previously unseen and coordinated attacks.

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